# NIRMAL KUMAR V.

B.Tech Information Technology • Kongu Engineering College

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**LeetCode**: @nirmal-kumar-v • **GeeksforGeeks**: @nirmalkumar33 • **Codeforces**: @nirmal_kumar_v

## EDUCATION

**Kongu Engineering College — Perundurai, Tamil Nadu** **2023 – Present**

Bachelor of Technology in Information Technology | **CGPA: 8.62 / 10.0** (Semesters 1–4) Current: 3rd Year / 5th Sem

HSC (Class XII): **91.33%** | **Computer Science: 100 / 100** March 2024

## TECHNICAL SKILLS

- **Languages:** Java (Primary for DSA & OOP), JavaScript (ES6+), Python, SQL
- **Web & Frontend:** React.js, Vite, HTML5, CSS3, Tailwind CSS, Responsive Design
- **Backend & APIs:** Node.js, Express.js, RESTful Architecture, JWT Authentication, FastAPI
- **Databases:** PostgreSQL, MongoDB, Relational Schema Design, Query Optimization
- **Developer Tools:** Git, GitHub, Linux CLI, Android / Termux Integration, Postman, Vercel

## PROBLEM SOLVING & COMPETITIVE PROGRAMMING

- **LeetCode: 250+ Problems Solved** across Data Structures, Dynamic Programming, and Graph Algorithms.
- **GeeksforGeeks: 500+ Problems Solved** (Coding Score: 1,500+; Institute Rank: 35; Longest Streak: 32).
- **Elite PAAC Training:** Selected among the **top ~4%** (80 students out of 2,000+ candidates considered) for specialized DSA and Competitive Programming training conducted directly by GeeksforGeeks mentors.

## SELECTED TECHNICAL PROJECTS

**Kongu Kural — Multilingual AI College Chatbot & SMS Gateway** **1st Prize — Tamil Mandram 24H Hackathon**

*Tech: React, Node.js, Express, NLP, Android Termux SMS Gateway, Vercel* | Live Demo • GitHub Role: Team Lead

- Architected a dual-channel conversational platform answering campus queries across Tamil, Tanglish, English, and Hindi using verified institutional knowledge bases.
- Engineered an Android Termux-powered SMS gateway with an offline fallback mechanism, enabling low-connectivity students to query database records via SMS.

**Integrated Retail POS & Store Management System** **1st Place — DevForge 2K26 24H Hackathon**

*Tech: React, Node.js, Express.js, PostgreSQL, Role-Based Access Control* | GitHub Role: Team Lead

- Led a 4-member sprint developing an offline-first retail system featuring client-side transaction queueing and deterministic background synchronization upon reconnection.
- Constructed relational PostgreSQL schemas, role-based authorization (Admin/Manager/Cashier), branch-level isolation, and real-time inventory decrement triggers.

**OptiVest / PortfolioIQ — Quantitative Portfolio Risk & Return Optimizer** **1st Prize — CoE Hackathon**

*Tech: React, Vite, FastAPI, Python, SciPy, Modern Portfolio Theory* | Live Demo • GitHub Role: Team Lead

- Implemented financial optimization endpoints in Python using SciPy to generate the Markowitz Efficient Frontier curve and validate asset covariance matrices.
- Ran Monte Carlo simulations across 10,000 randomized asset allocations to identify Maximum Sharpe and Minimum Volatility portfolio weights with dynamic visual charts.

**ChangeWave — Social Impact & Community Welfare Management System** **1st Place (RRC) • 2× 2nd Place (CSI)**

*Tech: Node.js, Express.js, MongoDB, JWT, JavaScript* | changewave.tech • GitHub Role: Presenter / Developer

- Developed an end-to-end welfare platform for campaign launch, pre/post initiative surveying, and sentiment metric tracking; presented as the institutional showcase system for NBA accreditation.

## TRAINING & EXPERIENCE

**In-Plant Trainee — Mist Technology (15 Days)** **Technical Training**

Completed 15 days of intensive training: 8 days in Full-Stack Web Development and 7 days in Frontend Application Design. Web & Frontend Engineering

**WeLe — Software Engineering Internship Selection:** Selected for an internship opportunity following technical evaluation at VIT Data Quest 3.0. Internship Selection

## CERTIFICATIONS & KEY HONORS

- **NPTEL (IIT) — Human-Computer Interaction: Elite + Gold Medal (Top 5% Nationwide, Score: 98%)**.
- **NPTEL (IIT) — Demystifying Networking:** Completed with 65% certification score.
- **Hackathon Recognition: 4 First-Place Wins** (DevForge 2K26 24H, Tamil Mandram 24H, CoE Challenge, RRC 8H) and **5 Second-Place Awards** across regional and national technical symposiums (CSI Solo & Team, IEEE BCI Paper, KPR Presentation).